

BSU Cyber Squad Camp 2026

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July 13 – 17, 2026

Monday – Friday | No prior experience needed



ARRIVAL / CHECK-IN

8:45 – 9:25 a.m.

CAMP HOURS

9:25 a.m. – 3:35 p.m.

PICK-UP

3:35 – 4:15 p.m.

LOCATION

Room 113B, CS Building

Buffet lunch provided daily

WHAT YOU'LL LEARN

1

Mon, Jul 13

Python basics and how the internet works

2

Tue, Jul 14

CIA triad, passwords, and account security

3

Wed, Jul 15

Cryptography: ciphers, hashing, and TLS

4

Thu, Jul 16

Web security: SQL injection, XSS, OWASP

5

Fri, Jul 17

CTF competitions, blue team, and careers

DAILY SCHEDULE

Time

Session

Slides

9:25 – 12:05

Lectures & Python exercises

Parts 1, 2, 3, 6

12:05 – 12:55

Lunch

12:55 – 1:45

Cyber Range labs

Part 4

1:45 – 2:35

Interactive game exercises

Parts 5, 7

2:35 – 3:25

Pair work & presentations

Part 9

3:25 – 3:35

Wrap-up

Part 8

WHAT YOU NEED

Laptop + Browser

Everything runs in the cloud - no software installation needed

Google Colab

Free Python notebooks - sign in with any Google account

US Cyber Range

Lab credentials provided on Day 1 for all five environments

Day 1

Python & Internet

Day 2

Passwords

Day 3

Cryptography

Day 4

Web & AI

Day 5

CTF & Blue Team

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SESSION DETAILS**Lectures (9:25 – 12:05)**

Get oriented and learn Python basics. Part 1: Orientation. Part 2: Linux, networking, CLI, SSH. Python exercise intro and Cyber Range walkthrough (Parts 3, 6).

Cyber Range (12:55 – 1:45)

Cyber Basics (2025.7) — VM login, CLI navigation, SSH into remote host.

Interactive Game (1:45 – 2:35)

Part 5 — Network packet simulation game. Students build and inspect packets.

Pair Work (2:35 – 3:25)

Day 1 coding challenge in Day1_student.ipynb. Work with your partner on CLI scripts.

Wrap Up (3:25 – 3:35)

Recap Linux and networking. Preview Day 2: passwords and hashing.

KEY CONCEPTS**Python**

Variables, loops, functions

CLI

Linux command-line navigation

SSH

Secure remote shell access

OSI Model

7-layer network stack

TCP/IP

Internet protocol suite

Subnetting

Dividing IP address space

Port

Numbered network endpoint

Firewall

Network traffic filter

Day 1

Python & Internet

Day 2

Passwords

Day 3

Cryptography

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Web & AI

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SESSION DETAILS**Lectures (9:25 – 12:05)**

Security foundations: CIA triad, threat actors, authentication (Part 2).
Password entropy math, keyspace calculation, Python exercise (Parts 3, 6).

Cyber Range (12:55 – 1:45)

Cyber Basics (2025.7) — Password cracking, hash comparison, salt demo.

Interactive Game (1:45 – 2:35)

Part 5 — Password strength game and threat scenario quiz.

Pair Work (2:35 – 3:25)

Build a keyspace calculator and hash cracker in Day2_student.ipynb.

Wrap Up (3:25 – 3:35)

Recap CIA triad and hashing. Preview Day 3: cryptography.

KEY CONCEPTS**CIA Triad**

Confidentiality,
Integrity, Availability

Hashing

One-way fingerprint
function

Salt

Random bytes added
before hashing

Entropy

Measure of password
randomness

Auth Factor

Something you
know/have/are

Threat Actor

Who carries out an
attack

Risk

Likelihood x Impact

AES

Advanced Encryption
Standard

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SESSION DETAILS

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History of cryptography and symmetric ciphers. Caesar, Base64, XOR (Part 2). RSA walkthrough, TLS handshake, frequency analysis, Python exercise (Parts 3, 6).

Cyber Range (12:55 – 1:45)

Kali Linux (2025.7) — Capture and decrypt traffic, frequency analysis challenge.

Interactive Game (1:45 – 2:35)

Part 5 — Cipher challenge game and frequency analysis puzzle.

Pair Work (2:35 – 3:25)

Implement Caesar cipher, XOR encoder, and Base64 decoder in `Day3_student.ipynb`.

Wrap Up (3:25 – 3:35)

Recap cipher types and TLS. Preview Day 4: web security.

KEY CONCEPTS

Caesar

Shift-based substitution cipher

Base64

Binary-to-text encoding

XOR

Bitwise exclusive-or cipher

RSA

Public-key asymmetric crypto

TLS

Transport Layer Security

Freq. Analysis

Breaking ciphers by letter count

Public Key

Freely shared encryption key

Certificate

Signed proof of identity

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Web app architecture, URLs, client-server model. SQL injection and XSS (Part 2). OWASP Top 10, session hijacking, AI prompt injection, Python exercise (Parts 3, 6).

Cyber Range (12:55 – 1:45)

Kali Linux (2025.7) — SQLi and XSS on a vulnerable web app, Burp Suite intro.

Interactive Game (1:45 – 2:35)

Part 5 — OWASP attack and defend simulation game.

Pair Work (2:35 – 3:25)

URL parser, input validator, and prompt injection guard in Day4_student.ipynb.

Wrap Up (3:25 – 3:35)

Recap OWASP Top 10 and AI attacks. Preview Day 5: CTF.

KEY CONCEPTS

SQL Injection

Malicious SQL in user input

XSS

Script injected into web page

OWASP Top 10

Most critical web vulnerabilities

Session Cookie

Token proving you are logged in

Burp Suite

Web security testing proxy

Prompt Inject.

Hijacking LLM instructions

WAF

Web Application Firewall

CSRF

Cross-Site Request Forgery

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Passwords**Day 3**
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Lectures (9:25 – 12:05)

CTF categories, OSINT methodology, blue team roles. Incidents, logs, OSINT (Part 2). Log analysis, cert roadmap, Python exercise, career paths (Parts 3, 6).

Cyber Range (12:55 – 1:45)

Kali Linux (2025.7) — Mini CTF: crypto, web, and forensics challenges.

Interactive Game (1:45 – 2:35)

Part 5 — Analyze login logs, identify the breach, write an incident report.

Presentations (2:35 – 3:25)

5 pairs, approximately 10 minutes each, or 10 pairs, approximately 5 minutes each: present your week project to the group.

Wrap Up (3:25 – 3:35)

Certificates, celebration, and cybersecurity career resources.

KEY CONCEPTS

CTF

Capture the Flag competition

OSINT

Open Source Intelligence

Blue Team

Defenders of a network

IOC

Indicator of Compromise

SIEM

Security event log aggregator

Forensics

Recovering digital evidence

Incident Resp.

Structured breach response

CompTIA Sec+

Entry cybersecurity certificate